



field guide · reading the body · 6 projects · jul 2026

> build six things that read the body_

Six EE Projects That Read The Body

ECG, EMG, EEG, and optical pulse – six builds that pull a living signal out of a room full of noise. From a \$15 sensor that finds your heartbeat in a beam of light to a robotic hand that closes when you flex, to catching an 8 Hz rhythm from your own visual cortex. Cheap parts; real precision analog.

01	Optical Pulse Sensor (PPG)	Shine light through your finger and watch your heartbeat appear	\$15
02	Biopotential Front End	The instrumentation amplifier that pulls a 1 mV signal out of a room full of mains hum	\$20
03	ECG Heart Monitor	Watch your own heartbeat draw itself on screen, wave by wave	\$25
04	EMG Muscle Sensor	Clench your fist and watch the electrical storm in your forearm	\$25
05	EMG-Controlled Robotic Hand	Flex your forearm and a robotic hand closes – myoelectric control, built on your bench	\$70
06	EEG Alpha-Wave Detector	Close your eyes and watch an 8 to 12 Hz rhythm rise out of the noise	\$60

● By Enes · @1nska · Electrical Engineer – a Build List field guide

Optical Pulse Sensor (PPG)

Shine light through your finger and watch your heartbeat appear

\$15

beginner

1 weekend

why this matters

The gentlest way into biosignals uses no electrodes at all. Shine light into tissue and a photodiode sees it dim slightly with every heartbeat, because each pulse of blood absorbs a little more. That is photoplethysmography (PPG) – the exact sensor on the back of every smartwatch and fitness ring. A \$5 module gives you a raw optical waveform; your job is to pull the tiny pulsatile component out of a big steady one, filter it, and count peaks into a heart rate. It teaches the AC-on-DC structure that runs through every biosignal in this guide, without ever touching a skin electrode.

WHAT THIS PROVES TO A RECRUITER

You understand photoplethysmography, separating a small AC signal riding on a large DC baseline, band-pass filtering, peak detection, and why motion artifacts dominate wearable sensing – the core of consumer health-device engineering.

→ the job this maps to: wearables and consumer health-device engineering – smartwatch, fitness-tracker, and ring companies.

! NOT A PULSE OXIMETER

This shows a pulse waveform and a heart rate. It is not a medical instrument. Real SpO₂ readings require empirical calibration against a clinical reference on actual subjects – a number your build cannot produce trustworthily. Never make a health decision from it, and if you are worried about your heart or breathing, see a doctor, not a breadboard.

how it works

- ▶ **Light in, blood out:** An LED sends light into the tissue and a photodiode measures how much comes back. Blood volume rises with each heartbeat and absorbs more light, so the detected signal dips in time with your pulse.
- ▶ **AC riding on DC:** Most of what the photodiode sees is constant – skin, bone, static blood. The heartbeat is a tiny fluctuation (the AC part) on top of that big DC level. Extracting a clean AC from a dominant DC is the whole game.
- ▶ **Why green for heart rate:** Green light is strongly absorbed by haemoglobin and penetrates shallowly, giving a strong pulse signal with less motion sensitivity. That is why fitness wearables glow green on your wrist.
- ▶ **Red and infrared for oxygenation:** Oxygenated and deoxygenated haemoglobin absorb red (~660 nm) and infrared (~940 nm) light differently. Comparing the pulsatile signal at both wavelengths is how commercial pulse oximeters estimate SpO₂ – but only after calibration against a clinical reference.
- ▶ **Band-pass and peak detect:** A heart rate lives around 0.7–3 Hz (40–180 bpm). Band-pass the signal into that window to kill DC drift and high-frequency noise, then detect peaks and time the intervals to get beats per minute.

bill of materials

component	part / spec	qty	~cost
PPG sensor module	MAX30102 (red + IR + photodiode)	1	\$5
Microcontroller	ESP32 / Arduino (I2C)	1	\$8
Display (optional)	OLED for live waveform	1	\$4
Passives + breadboard	wiring	–	\$2

build steps

1 Read the raw sensor

Wire the MAX30102 over I2C, load the library, and stream the raw red and IR counts. Rest a fingertip on it and confirm the numbers wobble slightly with each heartbeat.

2 Plot the waveform

Stream the raw values to a plot. You should see a large baseline with a small ripple – that ripple is your pulse. Note how much bigger the DC part is than the AC part.

3 Filter to the pulse band

Band-pass roughly 0.7–3 Hz to remove the DC baseline and high-frequency noise. The waveform should now show clean, repeating pulse peaks.

4 Count the beats

Detect peaks, measure the interval between them, and convert to beats per minute. Compare to your pulse counted by hand at the wrist to sanity-check the number.

5 Break it on purpose

Move your finger, press harder, press lighter, and watch the signal collapse into motion artifact. Understanding why wearables struggle during exercise is the real lesson here.

interview questions

- What physical effect does a PPG sensor actually measure?
- Why is the pulsatile part of a PPG signal so small compared to the baseline?
- Why do fitness wearables use green light for heart rate?
- How do red and infrared light let a pulse oximeter estimate oxygenation?
- Why can a DIY SpO₂ reading not be trusted without calibration?
- What causes motion artifacts in PPG, and how would you reduce them?

Biopotential Front End

The instrumentation amplifier that pulls a 1 mV signal out of a room full of mains hum

\$20

intermediate

1 weekend

why this matters

Your body produces electrical signals measured in millivolts – and it also acts as an antenna that picks up the mains hum flooding every room, often a thousand times larger than the signal you want. The instrumentation amplifier is the circuit that solves this: it amplifies only the difference between two electrodes while rejecting everything they see in common. Add a driven-reference electrode and careful filtering and suddenly a heartbeat or a muscle twitch emerges from the noise. This is the single circuit that makes ECG, EMG, and EEG possible, and building it yourself – rather than buying a module – is what separates people who understand biosignals from people who plug in a breakout board.

WHAT THIS PROVES TO A RECRUITER

You understand instrumentation-amplifier topology, common-mode rejection (CMRR), driven-reference (right-leg drive) circuits, electrode half-cell offsets and why you must AC-couple, gain staging, and mains-hum rejection – the foundation of all precision biomedical analog design.

→ **the job this maps to:** medical-device and precision-analog engineering – patient-monitoring, diagnostics, and analog front-end teams.

! BATTERY POWER, ALWAYS – AND NEVER INJECT CURRENT

Anything wired to your skin must run on batteries and must never be connected to mains-powered gear (a plugged-in laptop, a bench scope, a wall adapter) while electrodes are on a body. Use series resistors in every electrode lead, and isolate (Bluetooth, or an unplugged battery laptop) if you need to see the data on a computer. Everything in this guide only **measures** – never build anything that pushes current into a person, and do not attach electrodes to anyone with a pacemaker or implanted device.

how it works

- ▶ **The signal is tiny, the noise is huge:** An ECG is about 1 mV, an EMG a few mV, an EEG only tens of microvolts. Meanwhile your body couples capacitively to mains wiring and carries volts of 50 / 60 Hz interference. The signal is buried a thousandfold.
- ▶ **Differential rescue:** That interference appears almost identically on both measuring electrodes (it is common-mode), while the biosignal appears as a difference between them. An instrumentation amplifier amplifies the difference and rejects the common part – its ability to do so is its CMRR.
- ▶ **Driven reference (right-leg drive):** Better still, sense the common-mode voltage on the body, invert it, and feed it back through a third electrode. This actively cancels the interference at the source and buys you far more rejection than the amplifier alone – without ever tying the body to ground.
- ▶ **Electrode offsets force AC coupling:** Two Ag/AgCl electrodes on skin produce a DC half-cell voltage of up to a few hundred millivolts – hundreds of times bigger than the signal. If you applied full gain to that, the amplifier would slam into its rail. So you take modest gain first, high-pass to strip the DC, then apply the rest of the gain.
- ▶ **Filtering to the band you want:** Each biosignal has a band: ECG roughly 0.5–40 Hz, EMG roughly 20–450 Hz, EEG roughly 1–30 Hz. Band-pass to that window (plus an optional 50 / 60 Hz notch) and you throw away most of the remaining noise.

bill of materials

component	part / spec	qty	~cost
Instrumentation amp	INA333 or AD620	1	\$6
Op-amp	for driven reference + filters (quad)	1	\$2
Ag/AgCl electrodes	disposable snap electrodes (pack)	1	\$6
Passives + breadboard	resistors, caps, lead resistors	–	\$4
Batteries	9 V pair (isolated supply)	2	\$3

build steps

1 Build the amplifier core

Wire the instrumentation amp with a modest first-stage gain (about 10). Power it from batteries only. Test it with a signal generator or a divider before it ever touches skin.

2 Add series lead resistors

Put a current-limiting resistor (tens of k Ω) in each electrode lead. This is a safety measure, not an optional extra, and it costs you nothing in signal quality.

3 AC-couple and add gain

High-pass at roughly 0.5 Hz to strip the electrode DC offset, then add a second gain stage (100 or more) plus a low-pass for your band. Confirm the output no longer saturates.

4 Add the driven reference

Buffer and invert the common-mode voltage and drive it back into a third electrode on a bony spot (hip or elbow). Watch the mains hum on your output collapse – this is the most satisfying moment in the build.

5 Prove it on a real signal

Put two electrodes across your forearm, clench your fist, and watch the burst of muscle activity. You now have the front end that every remaining project in this guide depends on.

interview questions

- Why is an instrumentation amplifier used instead of a single op-amp for biopotentials?
- What is CMRR, and why does it dominate biosignal front-end design?
- How does a driven-reference (right-leg drive) circuit improve noise rejection?
- Why must a biopotential amplifier be AC-coupled before high gain?
- Why is the gain split across stages instead of applied all at once?
- Why must a body-connected circuit be battery powered and isolated?

ECG Heart Monitor

Watch your own heartbeat draw itself on screen, wave by wave

\$25

intermediate

1 weekend

why this matters

Every beat of your heart is a coordinated electrical wave, and it is large enough to read from your skin. Put two electrodes across your chest or arms, run them into a biopotential front end, and the classic PQRS complex appears on screen: the small P wave as the atria fire, the sharp QRS spike as the ventricles contract, the broad T wave as they reset. Then detect those R peaks and you have a live heart rate and beat-to-beat variability. It is the most iconic biosignal there is, the payoff is unmistakable, and it is the project that proves you can build precision analog that works on a living system.

WHAT THIS PROVES TO A RECRUITER

You understand cardiac electrophysiology at the signal level, lead configurations, the ECG bandwidth, R-peak detection algorithms, heart-rate and HRV extraction, and safe patient-connected design – the exact skill set of medical-device and patient-monitoring engineering.

→ **the job this maps to:** medical-device engineering – patient monitoring, cardiac diagnostics, and clinical-instrumentation teams.

! NOT A MEDICAL DEVICE

This is an engineering demonstration, not a diagnostic tool. It is not calibrated, not validated, and not certified. It can show you a waveform and a heart rate; it cannot tell you whether your heart is healthy, and a normal-looking trace means nothing clinically. Never make a health decision from it. If you have any cardiac concern, see a doctor.

how it works

- ▶ **The heart as a moving dipole:** As the cardiac muscle depolarises, it creates an electrical wavefront that behaves like a rotating dipole in your torso. Electrodes on the skin see the projection of that dipole – which is why where you place them changes what you see.
- ▶ **Reading the PQRS:** The P wave is atrial depolarisation, the QRS complex is the much larger ventricular depolarisation, and the T wave is ventricular repolarisation. Their shapes and timings are what clinicians read; for you, the tall R peak is the easiest landmark.
- ▶ **Leads and placement:** Einthoven's triangle defines the classic limb leads. Lead I runs right arm to left arm; Lead II (right arm to left leg) usually gives the biggest, cleanest R peak – a good first target. A third electrode carries the driven reference.
- ▶ **The right bandwidth:** Monitoring-grade ECG lives roughly between 0.5 and 40 Hz. Band-pass there and you remove electrode drift below and muscle noise above, without mangling the QRS shape.
- ▶ **Finding the beats:** The classic R-peak detector (Pan-Tompkins) band-passes the signal, differentiates it, squares it, integrates over a moving window, and thresholds the result. Time the intervals between R peaks and you have heart rate and beat-to-beat variability.

BUILD NOTE

The AD8232 packages the whole front end – instrumentation amp, right-leg drive, and filters – into one \$10 chip. Start there to get a waveform fast, then rebuild it with your project-2 front end to prove you understand what the chip is doing.

key design decisions

- **Why not just notch out the mains hum and skip the fancy front end?** A notch filter removes interference *after* the amplifier – but if the hum is a thousand times larger than the signal, it has already saturated your gain stages and the ECG is unrecoverable. And a notch near the QRS band distorts the waveform. The real fix is rejecting the interference at the input, with CMRR and a driven reference, so it never gets amplified in the first place.
- **Why drive a reference electrode instead of just grounding the body?** Tying a person to circuit ground creates a low-impedance path for fault current – a safety hazard – and it does not actually cancel the common-mode voltage. A driven-reference electrode feeds back an inverted copy of the common-mode signal through a high-value resistor, actively cancelling the hum while keeping the connection current-limited and safe.
- **Why AC-couple when the ECG has low-frequency content?** Electrode half-cell offsets reach hundreds of millivolts against a 1 mV signal, so without a high-pass the amplifier rails and you see nothing at all. The compromise is to set the corner low (about 0.5 Hz) – enough to strip electrode drift while preserving the ECG's slow features like the T wave.

bill of materials

component	part / spec	qty	~cost
ECG front end	AD8232 module (or your project-2 front end)	1	\$10
Ag/AgCl electrodes	disposable snap electrodes (pack)	1	\$6
Microcontroller	ESP32 / Arduino (ADC)	1	\$8
Battery	9 V or LiPo (isolated supply)	1	\$3
Passives	lead resistors, wiring	—	\$2

build steps

1 Wire it up on battery

Connect the front end and microcontroller and power the whole thing from a battery. Confirm it works before an electrode ever touches you, and keep it disconnected from any mains-powered device.

2 Place three electrodes

Two measuring electrodes (Lead I across the arms, or Lead II for a bigger R peak) and one driven reference on a bony landmark. Clean the skin so the contact is good – poor contact ruins everything downstream.

3 Get a raw trace

Stream the ADC values and plot them. Sit very still. You should see a repeating waveform with an unmistakable sharp spike – your QRS complex. Move and watch it drown in artifact.

4 Filter and clean up

Band-pass roughly 0.5–40 Hz and, if needed, notch the mains frequency. The P, QRS, and T features should now be clearly separable on the plot.

5 Detect beats and compute rate

Implement R-peak detection, compute beats per minute from the R-R intervals, and display it. Compare to your pulse counted by hand, then watch the rate rise after a set of push-ups.

interview questions

- What do the P wave, QRS complex, and T wave each represent physiologically?
- Why does electrode placement change the shape of the ECG you record?
- What is the appropriate bandwidth for a monitoring ECG, and why?
- Why is a notch filter alone insufficient to deal with mains interference?
- Describe how an R-peak detection algorithm works.
- What safety measures are mandatory for any patient-connected circuit?

EMG Muscle Sensor

Clench your fist and watch the electrical storm in your forearm

\$25

intermediate

1 weekend

why this matters

When you contract a muscle, hundreds of motor units fire and their summed action potentials reach the skin as surface EMG – a burst of noisy, high-frequency activity whose amplitude tracks how hard you are pushing. Reuse your biopotential front end, shift the filter band upward, rectify and smooth the result, and you get a clean “muscle effort” signal you can plot, threshold, or use as a control input. It is the biosignal most useful as an interface, and it is the direct prerequisite for the myoelectric hand in project 5.

WHAT THIS PROVES TO A RECRUITER

You understand surface EMG generation, why its band differs from ECG, envelope detection (rectify plus low-pass), electrode placement along muscle fibres, crosstalk between adjacent muscles, and turning a biosignal into a usable control value – the basis of human-machine interface work.

→ **the job this maps to:** biomedical sensing and human-machine interface engineering – neural-wearable, rehab-tech, and gesture-input teams.

how it works

- ▶ **Where the signal comes from:** A motor neuron fires and the muscle fibres it controls depolarise. Thousands of these overlapping action potentials sum into a noisy, spiky voltage at the skin – a few hundred microvolts to a few millivolts, much rougher-looking than an ECG.
- ▶ **A different band:** Most EMG energy sits between about 20 and 450 Hz. High-passing at ~20 Hz is essential: it rejects motion artifact and the ECG contamination that would otherwise pollute a recording taken anywhere near the torso.
- ▶ **The envelope is the useful part:** Raw EMG looks like noise, and its instantaneous value is meaningless for control. Rectify it (take the absolute value) and low-pass it, and you get a smooth envelope whose height tracks contraction strength – that is the signal you actually use.
- ▶ **Placement and crosstalk:** Put two electrodes on the belly of the target muscle, aligned with the fibre direction, with a reference on a bony landmark. Place them poorly and you will pick up neighbouring muscles – crosstalk, the main reason multi-gesture control is hard.
- ▶ **Fatigue and drift:** As a muscle tires, the EMG amplitude and frequency content shift, so a fixed threshold that worked ten minutes ago may not work now. Real systems recalibrate or normalise – a lesson that matters immediately in project 5.

bill of materials

component	part / spec	qty	~cost
EMG front end	your project-2 front end, or an EMG sensor module	1	\$12
Ag/AgCl electrodes	snap electrodes (pack)	1	\$6
Microcontroller	ESP32 / Arduino	1	\$8
Battery	isolated supply	1	\$3
Passives	filters, lead resistors	—	\$2

build steps

1 Retune the front end

Take your project-2 amplifier and move the band to roughly 20–450 Hz. Everything else – the instrumentation amp, driven reference, battery power – stays exactly as it was.

2 Place the electrodes

Two on the belly of your forearm flexor, aligned along the muscle, and a reference on your elbow. Clean the skin. Relax, then clench, and look for a burst of activity.

3 Look at raw EMG

Plot the raw signal. Relaxed, it should be near-silent; contracting, it should explode into noisy activity. That contrast is the whole signal.

4 Build the envelope

Rectify the signal and low-pass it (a few Hz) to produce a smooth envelope. Squeeze gently, then hard, and confirm the envelope height tracks your effort.

5 Turn it into a control

Map the envelope to something visible – an LED’s brightness, a bar graph, a servo angle. Set a threshold for “muscle on,” then check how well it holds up as your arm tires.

interview questions

- What physiological event produces a surface EMG signal?
- Why is EMG high-passed around 20 Hz, unlike ECG?
- What is envelope detection, and why is raw EMG unusable for control?
- How does electrode placement affect EMG quality and crosstalk?
- How does muscle fatigue change the EMG signal over time?
- How would you distinguish two different gestures from EMG?

EMG-Controlled Robotic Hand

Flex your forearm and a robotic hand closes – myoelectric control, built on your bench

\$70

advanced

2 weekends

why this matters

This is the build that makes people stop scrolling. Take the EMG envelope from project 4, map it to servo positions, and a tendon-driven robotic hand opens and closes in time with your own muscle – the same myoelectric principle behind real prosthetic limbs. Add a second EMG channel (flexor and extensor) and you can command open and close independently; map the envelope proportionally and the grip strength follows your effort. It ties every skill in this guide together into a single, visceral demo: biology in, mechanics out.

WHAT THIS PROVES TO A RECRUITER

You can take a noisy biological signal all the way through a real system – amplification, filtering, envelope extraction, calibration, thresholding, and actuator control – and make it work in real time on a living body. That end-to-end biosignal-to-actuator pipeline is exactly what prosthetics and neural-interface teams build.

→ **the job this maps to:** prosthetics, neural interfaces, and robotics – prosthetic-limb makers, neural-wearable companies, and rehab-robotics teams.

how it works

- ▶ **Envelope to angle:** The EMG envelope is a single number tracking contraction strength. Map it to a servo position and the finger closes as you squeeze harder – proportional myoelectric control, in one line of arithmetic.
- ▶ **Two channels, two commands:** One electrode pair on your flexor (closing) and another on your extensor (opening) gives you two independent inputs, so you can command open and close deliberately rather than relying on one muscle relaxing.
- ▶ **Calibration is mandatory:** Everyone's EMG amplitude differs, and it drifts with sweat, electrode position, and fatigue. Every session starts with a calibration: record a relaxed baseline and a maximum contraction, then normalise everything in between.
- ▶ **Tendon-driven mechanics:** Five servos pull fishing-line tendons through the fingers, with elastic returning them. The mechanics are simple and the compliance is forgiving – which is exactly why this architecture is so common in hobby and research hands alike.
- ▶ **What real prosthetics add:** Clinical myoelectric limbs use multi-channel EMG plus pattern-recognition machine learning to classify many grips, with socket fitting, robustness, and regulatory validation on top. Your build is the honest core of that idea, and the rest is engineering rigour.

BUILD NOTE

Don't design a hand from scratch. Print an open-source tendon-driven hand (InMoov and similar) or buy a servo hand kit, drive the fingers with fishing line, and power the servos from their own supply – not from the microcontroller's rail, which will brown out the moment five servos move at once.

bill of materials

component	part / spec	qty	~cost
EMG front end	project-4 sensor (×2 channels ideal)	1	\$15
Robotic hand	3D-printed tendon hand or servo hand kit	1	\$25
Servos	SG90 / MG90S (×5)	5	\$12
Microcontroller	ESP32 / Arduino	1	\$8
Servo power + battery	separate 5 V supply, isolated sensor battery	1	\$8
Electrodes + line	snap electrodes, fishing line, elastic	–	\$5

build steps

1 Build the hand

Print or assemble the tendon-driven hand, route fishing line through each finger to a servo horn, and add elastic returns. Test each finger by driving its servo directly – mechanics first, biology second.

2 Separate your power

Run the servos from their own 5 V supply with a common ground to the logic, and keep the EMG front end on its own isolated battery. Servo current spikes will wreck a shared rail and inject noise into your biosignal.

3 Feed in one EMG channel

Take the envelope from project 4 and map it to grip position. Clench and watch the hand close. Expect it to be jittery – that is the raw truth of biosignals.

4 Calibrate and smooth

Record your relaxed baseline and maximum contraction, normalise the envelope between them, add hysteresis so it does not chatter at the threshold, and smooth the servo motion. It should now feel deliberate.

5 Add the second channel

Add an extensor channel so one muscle closes and the other opens. Tune both, then film it – a hand that mirrors your own muscle in real time is the single best portfolio clip in this guide.

interview questions

- How does a myoelectric prosthesis translate muscle activity into motion?
- Why must an EMG control system be calibrated per session, per user?
- Why use separate flexor and extensor channels instead of one muscle?
- Why must servo power be separated from the biosignal front end?
- What is hysteresis, and why does a threshold-based control need it?
- How do clinical prosthetics classify multiple grips from EMG?

EEG Alpha-Wave Detector

Close your eyes and watch an 8 to 12 Hz rhythm rise out of the noise

\$60

advanced

2 weekends

why this matters

EEG is the hardest biosignal there is: tens of microvolts, a hundred times smaller than an ECG, buried under mains hum, muscle tension, and eye movement. But there is one effect a careful DIY rig can genuinely capture – the alpha rhythm. Relax with your eyes closed and an 8 to 12 Hz oscillation appears over the back of your head; open your eyes and it vanishes within a second. Watching that band power rise and fall on an FFT, on command, is the most demanding and most rewarding build in this guide – and it is a real, reproducible neuroscience result, not a gimmick.

WHAT THIS PROVES TO A RECRUITER

You can design a genuinely low-noise, high-gain analog front end, manage shielding and grounding at the microvolt level, distinguish real neural signal from muscle and eye artifact, and validate a result experimentally. Microvolt-level analog design and honest artifact rejection are precisely what neurotech teams look for.

→ **the job this maps to:** neurotech and brain-computer interface engineering – BCI startups, neural-wearable companies, and neuroscience research labs.

! IT WILL NOT READ YOUR MIND

Alpha blocking – a rhythm that appears when your eyes close and disappears when they open – is the realistic and honest goal. Anything claiming to read thoughts or emotions from a few electrodes is overselling. Most hobby “brainwave” projects are actually detecting jaw clenching or eye blinks; the eyes-closed alpha test is how you prove yours is real. This is also not a diagnostic device, and it cannot tell you anything about your neurological health.

how it works

- ▶ **Microvolts, not millivolts:** Scalp EEG is roughly 10–100 μV – ten to a hundred times smaller than an ECG. That means much higher gain (often 10,000 $\times$ or more), far lower noise, and no tolerance for the sloppy wiring you got away with earlier.
- ▶ **The alpha rhythm:** When a relaxed person closes their eyes, neurons in the visual cortex synchronise into an 8–12 Hz oscillation that is strongest over the occipital region, at the back of the head. Opening the eyes suppresses it almost immediately. This alpha blocking is the classic, reproducible EEG demonstration.
- ▶ **Electrodes and placement:** Measure over the occipital area, referenced to an earlobe or mastoid, with a driven reference electrode. Gold cup electrodes with conductive paste (parted through the hair) give far better contact than dry electrodes – and contact quality dominates your result.
- ▶ **Artifacts everywhere:** Eye blinks produce huge slow deflections, jaw and neck tension produce broadband muscle noise, and any cable movement produces artifact. All of these are bigger than the alpha you are chasing, which is why so many “brainwave” demos are really artifact detectors.
- ▶ **Proving it with an FFT:** Band-pass roughly 1–30 Hz, run an FFT, and watch the power in the 8–12 Hz bin. Eyes closed and relaxed: it rises. Eyes open: it drops. Toggling that on command, repeatably, is your proof the signal is neural and not noise.

BUILD NOTE

This is the one build where sloppiness guarantees failure. Battery power, shielded leads, short wires, a driven reference, and real electrode contact (gold cup electrodes with conductive paste, parted through the hair) are not optional extras – they are the difference between alpha waves and a screen full of mains hum.

bill of materials

component	part / spec	qty	~cost
Low-noise front end	INA333 (or ADS1299 for the serious build)	1	\$8
Precision op-amps	filters + driven reference	1	\$5
Gold cup electrodes + paste	occipital, reference, ground	1	\$20
Shielded cable + enclosure	short shielded leads, metal box	–	\$8
ADC / microcontroller	24-bit ADC or ESP32 + care	1	\$10
Batteries + passives	isolated supply, filters	–	\$9

build steps

1 Build for noise, not speed

Assemble the front end with very high gain, split across stages, band-passed to about 1–30 Hz with a mains notch. Keep leads short and shielded and run everything from batteries in a shielded enclosure.

2 Prove the rig on ECG first

Before you go near your head, record an ECG with the same rig. If you cannot get a clean ECG, you will never see EEG – this step will save you a weekend of confusion.

3 Place occipital electrodes

Part the hair and apply gold cup electrodes with paste over the occipital region, reference on the earlobe or mastoid, plus the driven reference. Check contact impedance; poor contact is the number one failure.

4 Sit still, close your eyes

Relax, stay very still, and record with eyes closed for thirty seconds, then eyes open for thirty seconds. Stillness matters enormously – any jaw tension will swamp the signal.

5 Find the alpha in the FFT

Run an FFT and compare the 8–12 Hz band power between the two conditions. A clear rise with eyes closed, repeatable across trials, is real alpha. Anything that only appears when you blink or clench is artifact – be honest about which you got.

interview questions

- Why is EEG so much harder to measure than ECG?
- What is the alpha rhythm, and what causes alpha blocking?
- Why is electrode contact impedance so critical in EEG?
- What are the dominant artifacts in EEG, and how do you identify them?
- How would you prove that a signal you recorded is neural and not muscle artifact?
- What are the realistic limits of a low-channel-count EEG system?

> wrap

Your body is broadcasting – most engineers just can't hear it.

A pulse in a beam of light, a heartbeat across your chest, a muscle firing, a rhythm in your visual cortex – six ways to pull a living signal out of noise a thousand times larger. The parts are cheap (a \$5 optical sensor, a \$6 instrumentation amp, a pack of electrodes). The skill is entirely in the analog: the common-mode rejection, the gain staging, the filtering, the honest separation of signal from artifact. That is exactly what medical-device, neurotech, and wearable teams hire for.

// the skill ladder

Build them in order and each one hands you the technique the next one needs:

optical pulse → biopotential front end → ECG → EMG → myoelectric control → EEG

! THE RULES, EVERY SINGLE TIME

Battery power only. Never connect a body-attached circuit to mains-powered equipment – no plugged-in laptop, no bench scope, no wall adapter. Put current-limiting resistors in every electrode lead, and isolate (Bluetooth, or a laptop running on battery) if you need the data on a computer. Everything here only **measures**; never build anything that pushes current into a person, and never attach electrodes to anyone with a pacemaker or implanted device. And none of these are medical devices – they are demonstrations, and no health decision should ever come from one.

DO THIS

Build even one. Capture the trace – a PQRS complex, an EMG envelope tracking your grip, a hand closing when you flex, alpha power rising when your eyes close – and put the code on GitHub with your filter design, your calibration, and an honest note on artifacts and scope. One biosignal demo and one clean repo prove you can do precision analog on a living system.

Full build notes, parts links, and the next field guide go out in **The Build List**. The people actually shipping these – and trading electrode and shielding tricks – are in **Ohm's Club**.

– Enes (@1nska) · Electrical Engineer